

SPECIAL EVENTS

PHASES OF THE MOON

	FULL MOON	NEW MOON
2020	September 2	September 17
2021	September 20	September 7
2022	September 10	September 25
2023	September 29	September 15
2024	September 18	September 3
2025	September 7	September 21
2026	September 26	September 11
2027	September 16	September 30

PLANETS

2021: September 14 Mercury is at greatest evening elongation, magnitude 0.4.

2022: September 26 Jupiter is at opposition, magnitude -2.9.

2023: September 2 Mercury is at greatest morning elongation, magnitude -0.2.

2024: September 5 Mercury is at greatest morning elongation, magnitude 0.0.

2024: September 8 Saturn is at opposition, magnitude 0.6.

2025: September 21 Saturn is at opposition, magnitude 0.6.

2027: September 24 Mercury is at greatest evening elongation, magnitude 0.3.

ECLIPSES

2024: September 18 A partial eclipse of the Moon is visible from North and South America, Europe, and Africa.

2025: September 7 A total eclipse of the Moon is visible from Europe, Africa, Asia, and Australia.

2025: September 21 A partial eclipse of the Sun is visible from the south Pacific, New Zealand, and Antarctica.

SEPTEMBER

Northern nights grow longer as the Sun approaches the celestial equator, but in the Southern Hemisphere, the nights shorten. On September 22–23, the Sun lies on the celestial equator, and day and night are of equal length worldwide. The rich band of constellations along the Milky Way, from Cygnus in the north to Sagittarius and Scorpius in the south, begins to give way this month to fainter constellations, many of them with watery associations, such as Capricornus, Aquarius, and Pisces.

NORTHERN LATITUDES

THE STARS

Cepheus, high up in the north, is best placed for evening observation this month and next. Its most celebrated star is Delta (δ) Cephei, the prototype of a class of pulsating variables. Deneb in Cygnus, Vega (see p.253) in Lyra, and Altair (see p.252) in Aquila—the stars of the Summer Triangle—remain high in the western half of the sky, while the Square of Pegasus is high in the east with Cassiopeia between it and the north celestial pole. The bright star Fomalhaut (see p.253) in

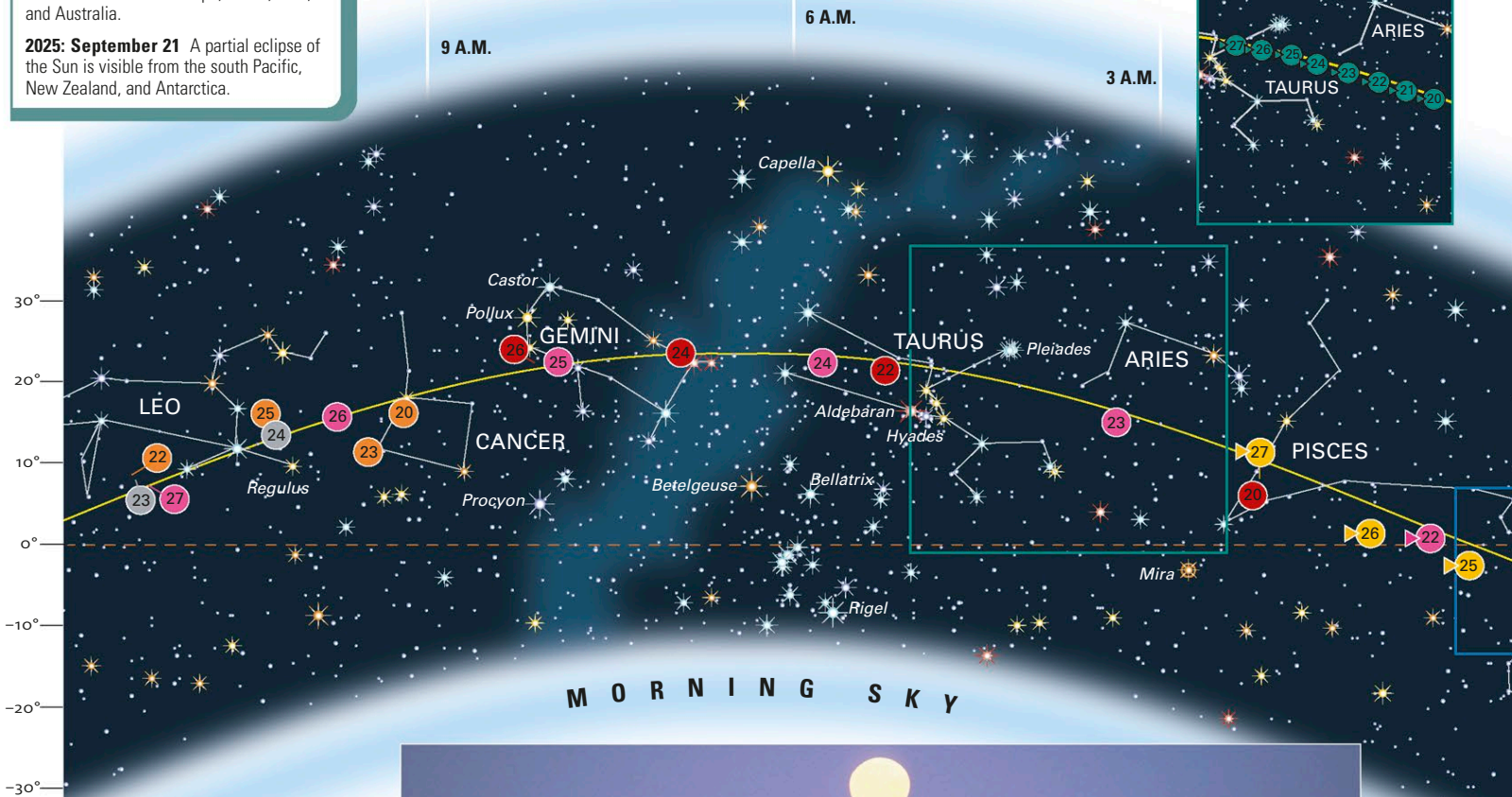
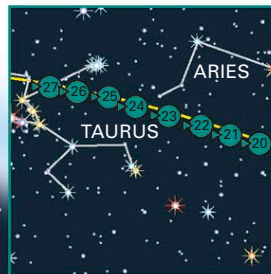
Piscis Austrinus is low in the south with Aquarius above it. A cascade of faint stars suggests the flow of water from the water carrier's urn toward the southern fish, Piscis Austrinus. For observers at high northern latitudes, this is the best time of year to see the zodiacal constellation Capricornus in the evening sky, lying low in the south to the right of Fomalhaut.

DEEP-SKY OBJECTS

Near Deneb in Cygnus lies one of the most distinctive nebulae in the sky, NGC 7000, popularly called

the North America Nebula, on account of its shape. Under clear, dark skies, it can be detected with binoculars, but it is best seen on long-exposure photographs. Another object of note in Cygnus is the open star cluster M39, which is visible through binoculars. The 6th-magnitude globular cluster M15, also visible through binoculars, is not far from the star Enif—Epsilon (ε) Pegasi—which marks the horse's nose in Pegasus.

URANUS



THE HARVEST MOON

The full moon that occurs closest to the northern fall equinox is termed the Harvest Moon because its light was said to assist farmers working late in the fields.

